

Matías Rodríguez Cárdenas

[LinkedIn](#) | [Github](#) | [Portfolio](#) | [Email](#)

ABOUT ME

Systems Engineer and Full Stack Developer with extensive experience in creating scalable software solutions. I combine a strong foundation in backend and frontend architectures with advanced expertise in Artificial Intelligence (GenAI, RAG, Autonomous Agents) and Data Engineering (ETL, MLOps). Passionate about transforming complex data into secure, high-performance enterprise products with a product-oriented mindset. C1 Advanced English, excelling in global, multidisciplinary teams.

SKILLS

Skills: AI & Data: LangChain, LangGraph, RAG, OpenAI API, Gemini 1.5 Pro, Scikit-learn, Airflow, Pinecone, ChromaDB.

Backend: Python (FastAPI), Java (Spring Boot), C, RESTful APIs, Redis, Server-Sent Events (SSE), Tool Calling.

Frontend: React 19, Next.js 16, TypeScript, Tailwind CSS v4, Vite, AG Grid, Recharts.

DevOps & DBs: PostgreSQL (RLS, Materialized Views), MySQL, Docker, AWS (S3, CloudFront), Clerk, MLOps.

EXPERIENCE

Full Stack Developer

Abaco Informatica

Tandil, Argentina

2021 - Present

- Spearheaded the project's migration from Create React App (CRA) to Vite, resolving deep dependency conflicts, updating obsolete packages, and implementing HMR to drastically enhance development performance.
- Designed and implemented a database optimization strategy using Materialized Views in PostgreSQL, which significantly reduced complex analytical query times without blocking transactional operations.
- Replaced legacy DataTables with AG Grid, engineering a client-side parser to auto-generate interactive grids from raw HTML and implementing advanced enterprise features (dynamic filters, multi-column sorting, PDF/Excel export suite).
- Architected a custom "DebugView" system, enabling real-time backend log viewing and an integrated SQL console directly from the frontend UI, removing the development team's need for server SSH access.

Full Stack Developer Intern

Cleatus AI

Remote

2025

- Led a comprehensive UI/UX overhaul of the GovCon AI conversational assistant, significantly improving usability and modernizing its visual aesthetic, leading to increased user retention.
- Designed and implemented interactive analytical dashboards to visualize key government procurement data and AI user engagement metrics.
- Executed strategic database schema modifications and refactoring to support high-throughput aggregations, optimizing data retrieval for real-time frontend dashboards.

PROJECTS

Fluent AI - Autonomous Multi-Agent Data Platform

Engineered a high-performance generative BI platform that transforms raw data into insights via a conversational interface.

- Agentic AI: Orchestrated a stateful multi-agent system using LangGraph (AsyncRedisSaver) and Gemini 1.5 Pro with autonomous tool-calling (dynamic SQL execution, vector search, chart config generation).
- Security & Architecture: Implemented absolute data isolation by injecting search_path at the connection level and enforcing PostgreSQL Row-Level Security (RLS) to prevent cross-tenant LLM hallucinations.
- Streaming: Developed a real-time chat pipeline utilizing Server-Sent Events (SSE) in an asynchronous FastAPI backend, seamlessly paired with a Next.js 16 and React 19 frontend.

AutoBid AI - Hybrid AI SaaS (Predictive + Generative)

Architected a B2B SaaS platform to analyze tenders and automate proposal writing.

- Machine Learning: Trained a Random Forest model and integrated SHAP values for explainable AI predictions on win probability.
- RAG Pipeline: Implemented multi-tenant semantic search with Pinecone to generate contextualized responses.
- Security: Integrated Presidio for automatic PII/sensitive data anonymization before LLM API processing.

AI-Powered K-Means Clustering App (Generative BI)

Developed a full-stack data analysis platform fusing classic ML with modern GenAI.

- Algorithm: Built a K-Means engine optimizing cluster count via the mathematical Elbow Method.
- Semantic Interpretation: Sent statistical heuristics to Gemini 1.5 Flash (protecting raw user data) to automatically generate business-friendly segment names and marketing strategies, visualized with interactive Recharts.

Chatbot with Automated ETL Pipeline (RAG + Data Eng)

Built an end-to-end “Chat with your CV” RAG assistant using FastAPI, Google Gemini, and ChromaDB.

- Data Engineering: Architected a robust, automated ETL pipeline using Apache Airflow and Docker that monitors a GitHub repository and autonomously updates the Vector DB upon detecting PDF changes.

Absenteeism Prediction MLOps Project

Implemented an end-to-end MLOps system to predict workplace absenteeism.

- MLOps Pipeline: Designed a decoupled architecture (Next.js, FastAPI, Vercel Postgres) featuring an automated Airflow retraining pipeline triggered by data ingestion events, comparing new model performance against production before deployment.

AI Technical Interviewer

Developed an AI-powered HR platform. The app gathers candidate data and generates dynamic questions by combining database queries with LLM-generated prompts. It accepts text/voice responses and uses AI to evaluate answer quality.

EDUCATION & CERTIFICATIONS

Google, Online	2026
<i>Google Professional AI Certificate</i>	
AI Engineer Bootcamp (Udemy), Online	2025
<i>LLMs, LangChain, Autonomous Agents Design</i>	
Data Engineering Bootcamp (Meta Engineers), Online	2025
<i>Airflow, ETL/ELT, Cloud Data Warehousing, Advanced SQL</i>	
Developer Boost Seminar (Tandil Tech Cluster), Tandil, Argentina	2025
<i>Generative AI for SDLC (Agents, MCP, Copilot, Cursor)</i>	
The Data Science Course 2024, Online	2024
<i>Calculus, Linear Algebra, Advanced Stats, TensorFlow</i>	
Bootcamps: Java Backend & Frontend Web Dev, Online	2024
<i>Spring Boot, Hexagonal Architecture, React.js</i>	
Universidad Nacional del Centro de la Provincia de Buenos Aires, Tandil, Argentina	2020-2024
<i>Systems Engineering</i>	
EF SET, Online	2024
<i>C1 - Advanced English Certificate</i>	